

Rhea Pandita

☎ (310)734-9846 | ✉ rheasuneelpandita@gmail.com | 🔗 linkedin.com/in/rhea-pandita | 🌐 rp-1106.github.io | 📄 github.com/RP-1106

EXPERIENCE

USC CeNEC A-Z Lab | Graduate Research Assistant

Aug 2025–Present | Los Angeles, CA

- Built a rate-limited LLM inference pipeline (deque-based scheduler + retry/backoff) enabling reliable batch evaluation across **GPT-OSS-120B, Claude-Opus-4.x, LLaMA-3.3-70B, and Qwen-32B**.
- Architected a modular evaluation framework (prompt orchestration, API parsing, structured logging), achieving **98% output extraction** across 158 annotated multimodal samples.
- Developed automated logging system (Python, Pandas) tracking per-sample outputs and aggregate metrics for reproducible cross-model benchmarking at scale.

KPIT Technologies | Machine Learning Intern

Jun 2024–Aug 2024 | Bangalore, India

- Engineered a Python-based **CAD processing pipeline** to parse IGES/STEP files and extract structured geometry for automated backend modification of 3D designs.
- Built geometric reconstruction algorithms mapping 2D shapes to 3D counterparts, enabling scalable CAD parameter editing across multiple files.

KPIT Technologies | Machine Learning Intern

May 2023–Aug 2023 | Bangalore, India

- Built a text-to-image generation system using **Stable Diffusion v1-4** (HuggingFace Diffusers), leveraging **CLIP text encoder, latent U-Net, and VAE** to generate automotive design concepts from natural language prompts.
- Developed an **image-to-3D pipeline** using **GLPN (vinvino02/glpn-nyu)** (monocular depth estimation) and **Open3D** (Poisson reconstruction) to output **.obj meshes** for rapid prototyping.

SKILLS

Languages: Python, SQL, C, JavaScript, TypeScript

AI/ML: LLMs, RAG, Transformers, HuggingFace, LangChain, FAISS, RAGAS

Libraries: PyTorch, Tensorflow, Pandas, NumPy, Scikit-learn, Matplotlib, NLTK, SpaCy

Frameworks/Tools: React, Node.js, Flask, Git

Databases: MySQL

Courses: Deep Learning, Reinforcement Learning, Natural Language Processing (NLP), Machine Learning, Statistical Inference, Data Mining, Big Data Analysis, Database Systems, Information Retrieval & Web Search Engines

PROJECTS

Engagement-Aware Movie Recommendation System 🌀 | Python, PyTorch, BERT4Rec, SASRec, LightGCN

- Achieved **0.3085 HR@10** on **BERT4Rec**, **0.2849 HR@10** and **0.1634 MRR** on **SASRec** with gated item-feature fusion and popularity-debiased loss on **72k users** and **649k interactions**.
- Improved **GRU-LightGCN** baseline by **21.5% HR@10** and **31.7% NDCG@10** via a lightweight **9.9K-parameter reranker** over top-50 candidates with zero catastrophic forgetting.

ICU Deterioration Prediction 🌀 | Python, PyTorch, ClinicalBERT, BioViL-T, Scikit-learn, Pandas, SHAP

- Prevented **0.15-0.20 AUROC inflation** by designing a **feature-cutoff** leakage barrier across **91k stays** and **35m clinical records**.
- Achieved **AUROC 0.9998 (89% recall)** on respiratory failure by fine-tuning **ClinicalBERT** on serialized patient notes with strict patient-level splits and val-guided epoch selection.
- Ran a **7-condition ablation** across modality combinations; identified tabular+chest X-ray features as optimal (**AUROC 0.8700**) by fusing structured data with **PCA-compressed BioViL-T** image embeddings.
- Built a **Streamlit** clinical decision support UI with **SHAP** feature attribution and colour-coded risk gauge for interpretable bedside risk assessment.

AI-Assisted Personal Finance Management System 🌀 | Python, LangChain, Groq API, ChromaDB, Ragas, MySQL

- Developed financial chatbots leveraging **Mistral-Saba-24B** via Groq API and Ollama, handling **4,530 transactions** with hybrid SQL and document querying.
- Built a **RAG pipeline** (LangChain, ChromaDB, HuggingFace Embeddings), reducing response latency **81%** and memory usage **33%** vs. OllamaLLM Mistral baseline.
- Curated a **330 question–SQL dataset** for prompt refinement; evaluated via **RAGAS** (faithfulness **1.0**, relevance **0.872**, responses **89% faster** than baseline).

EDUCATION

University of Southern California

Master of Science in Computer Science

Los Angeles, CA

Aug 2025–May 2027

Shiv Nadar University

B.Tech in Artificial Intelligence & Data Science

Chennai, India

Sep 2021–Jul 2025

PUBLICATION

- T.R. Althi, V. Kothamachu, **R. Pandita**, "Accelerating Automotive Design: Harnessing AI Models for Efficient 3D Design and Development of Automobile Systems and Subsystems" *IEEE ITEC-India*, 2023. doi:10.1109/ITEC-India59098.2023.10471378